

Supporting Information

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Fine Tuning the Optical Properties of Single Au Nanoparticles by Plasmon-Driven Growth in Closed-Loop Control

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Supporting Information

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S1. Size distribution of the 66 nm Au NPs seeds

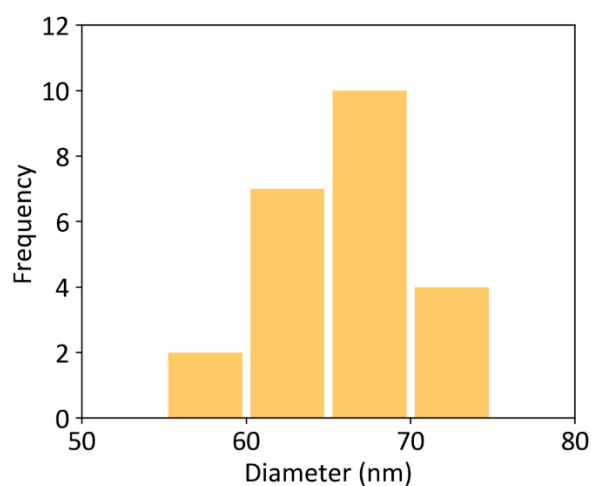


Figure S1. Histogram of diameter (nm) for 23 different Au NPs used as seeds, determined from scanning electron microscopy images. The average diameter is 66 nm and the standard deviation 4 nm.

S2. Error in the determination of λ_{max}^{PL}

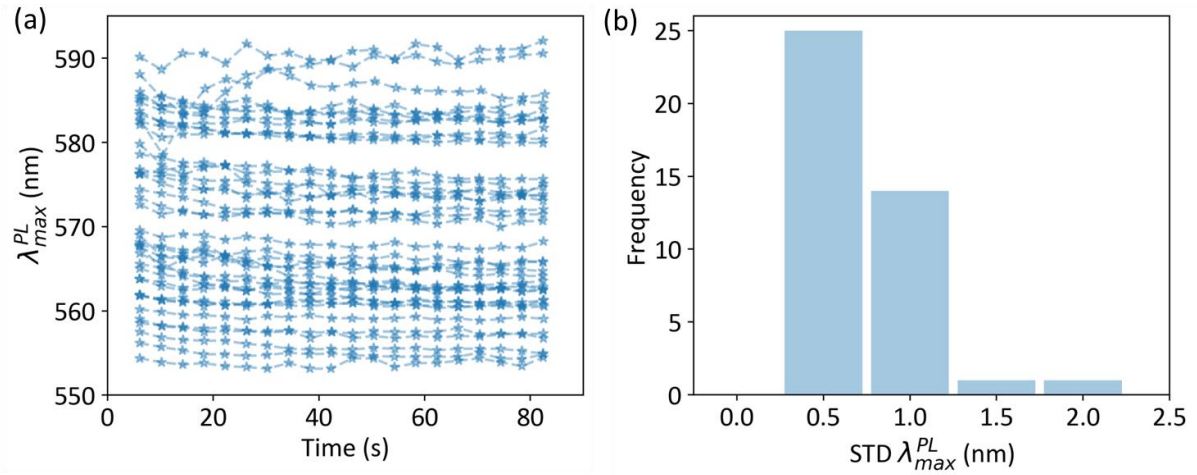


Figure S2. (a) λ_{max}^{PL} (nm) vs. time for 41 different Au NPs measured during 80 s on water with a laser irradiance of $2.6 \text{ mW} \cdot \mu\text{m}^{-2}$. The PL was measured for Au NPs with a λ_{max}^{PL} (nm) > 550 nm. (b) Histogram of the standard deviation of λ_{max}^{PL} on time ($\text{STD } \lambda_{max}^{PL}$ (nm)) for 41 Au NPs. The mean value of $\text{STD } \lambda_{max}^{PL}$ was 1 nm.

S3. Kinetics of the growth reaction.

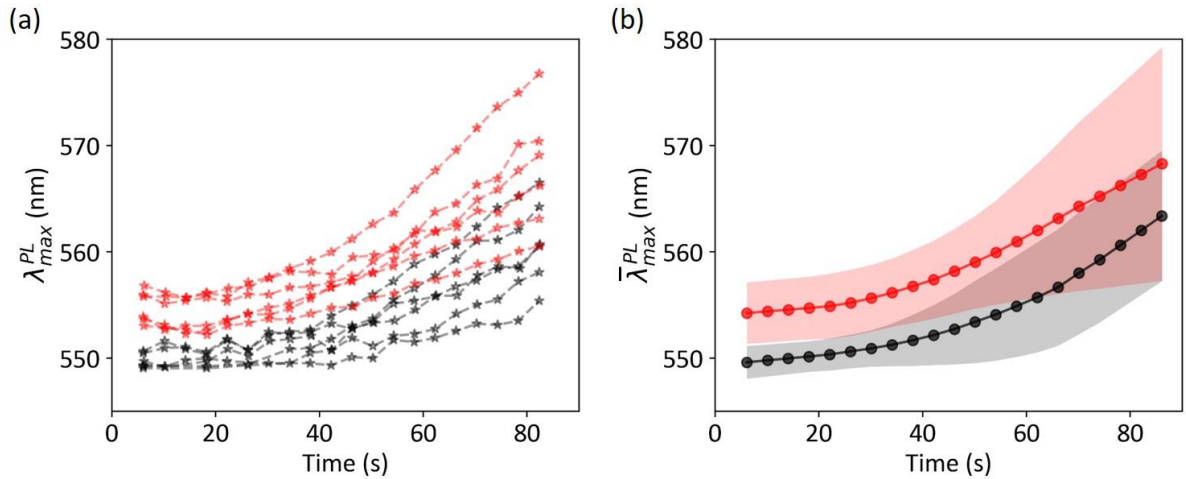


Figure S3. (a) Maximum of PL emission λ_{max}^{PL} vs. time for 12 Au NPs grown during 80 s under identical conditions: irradiance = $2.6 \text{ mW} \cdot \mu\text{m}^{-2}$, $0.5 \mu\text{M}$ HAuCl_4 . Black color corresponds to 6 NPs with $549 \text{ nm} < \text{initial } \lambda_{max}^{PL} < 551 \text{ nm}$. Red color corresponds to 6 NPs with $553 < \text{initial } \lambda_{max}^{PL} < 557 \text{ nm}$. (b) Mean $\bar{\lambda}_{max}^{PL}$ vs. time during plasmon-driven growth. Each color curve corresponds to the average kinetics and shaded areas indicate ± 2 standard deviations. Black curve has an initial standard deviation of 0.8 nm and final of 3.1 nm. Red curve has an initial standard deviation of 1.5 nm and final of 5.5 nm.

S4. Scattering spectra of the 66 nm Au NPs seeds

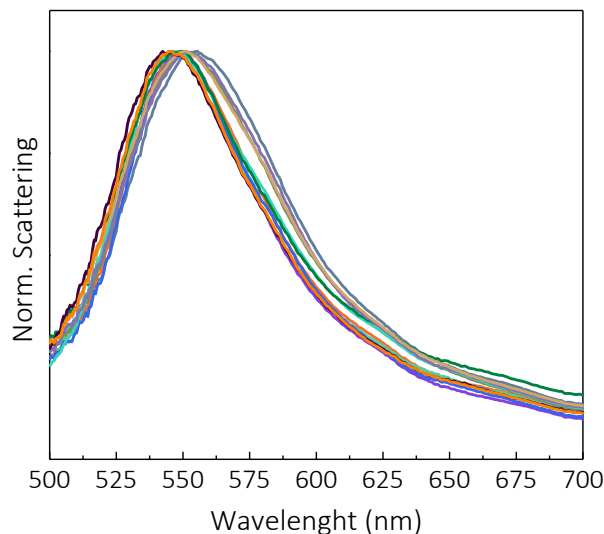


Figure S4. Normalized scattering spectra of 12 different Au NPs used as seeds, where the different maximums account for the size distribution.

S5. Histograms of λ_{max}^S of Au NPs seeds and plasmon-induced growth Au NPs

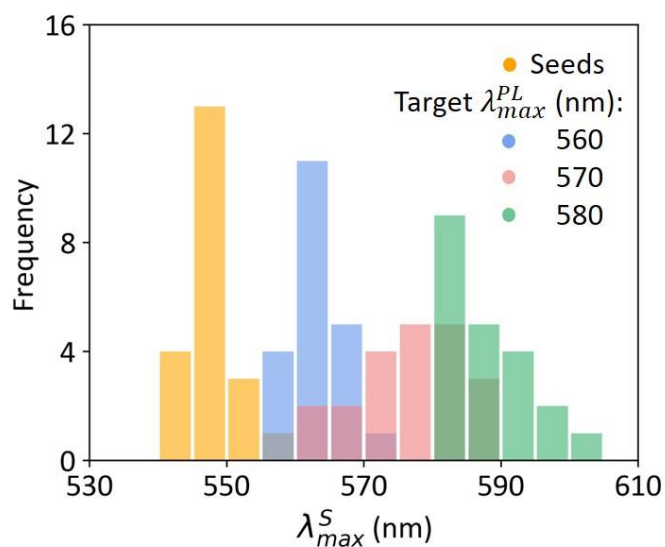


Figure S5. Histogram of λ_{max}^S of 21 different Au NPs before plasmon-driven growth (seeds) and grown to three different λ_{max}^{PL} targets: 560, 570 and 580 nm. The means and standard deviations of the λ_{max}^S distributions are: (547 ± 3) nm for the seeds, (563 ± 4) nm for the NPs with target λ_{max}^{PL} of 560 nm, (576 ± 7) nm for the NPs with target λ_{max}^{PL} of 570 nm, and (588 ± 7) nm for the NPs with target λ_{max}^{PL} of 580 nm.